Morality in Minds, Brains, and Machines

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Agenda

1. Unwrapping the **moral** mind with AI
2. How **AI** learns right from wrong
3. **Societal implications** of moral AI

For each of the following scenarios, please quickly indicate how **morally wrong** you personally find the action.

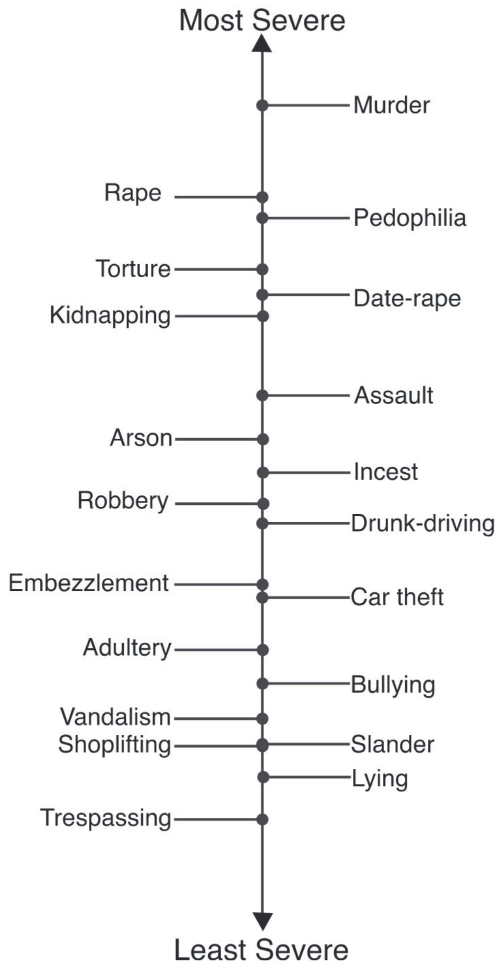
On human morality

Humans **universally** condemn what they see as moral transgressions

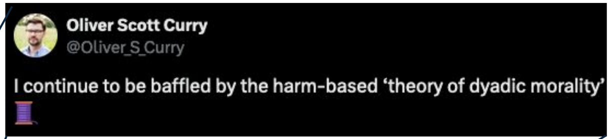
Graded wrongness of distinct violations **varies** across individuals, neurodiverse populations, and cultures

How the mind processes, organizes, and translates moral violations into subjective moral judgments remains an ongoing puzzle

⇒ **Machine learning and AI can help us resolve this puzzle**



The most active debate in moral psychology



Moral monists

Moral pluralists

Morality is unified by a single "essence" (e.g., harm)

Morality is too diverse to be unified (e.g., fairness, loyalty, …)

Defining morality objectively with machine learning

Machine learning is fundamentally about induction

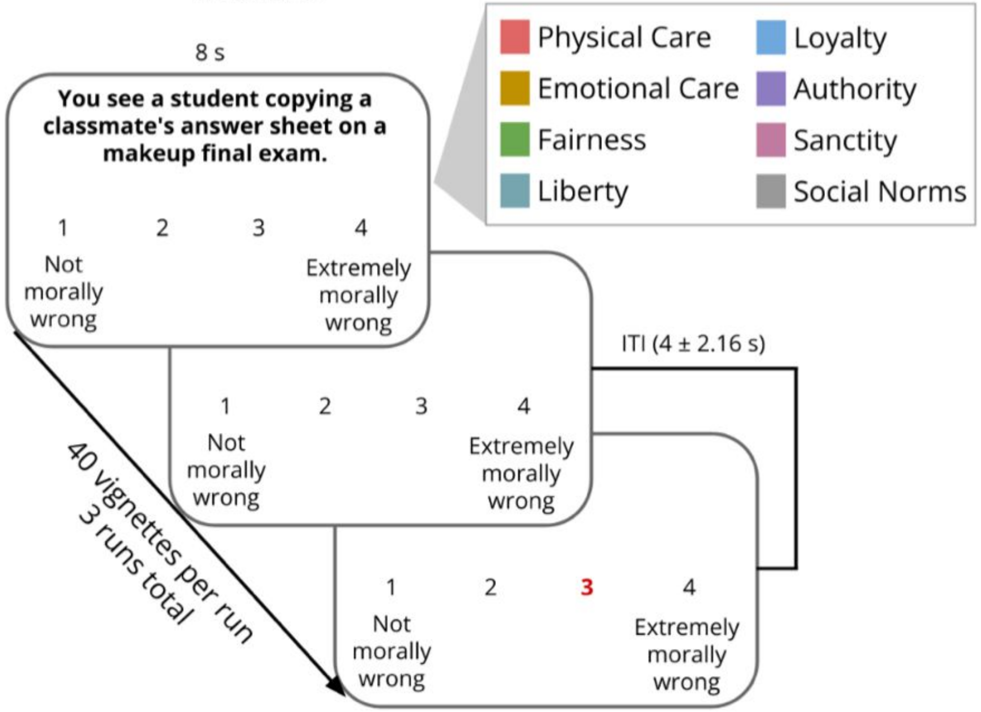
- The goal is to learn generalizable **rules** from a sample of observations
  (e.g., does intentional agent harm vulnerable patient?)

- For classification, this involves specifying a **decision boundary** based on a set of features that **categorizes** observations
  (can model accurately distinguish moral violations (pluralism) or does it get confused (monism) ?)

- The performance of a model in independent data provides evidence of **generalization**
  (can model generalize across individuals within and across cultures?)

Developing brain signatures of moral transgressions

Moral judgment paradigm



N = 64
120 vignettes
7,680 average “brain maps”
ca. 200k data points / brain map

Based only on brain imaging data, can a machine learning classifier (‘neural decoder’) predict which type of moral violation a person is currently judging?

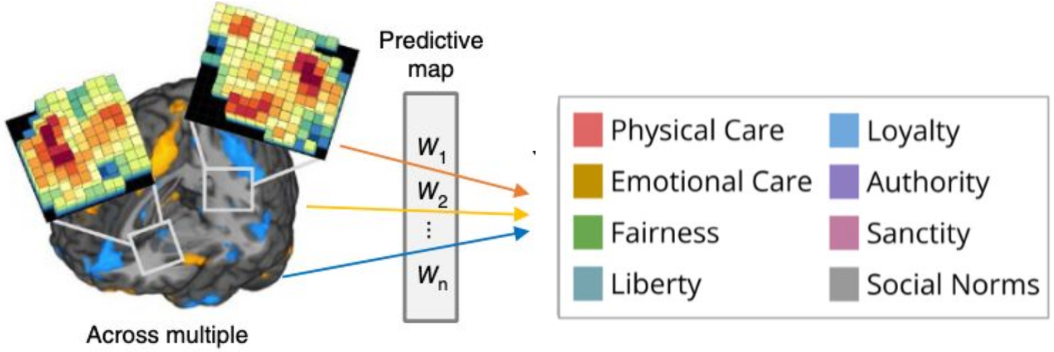
X (brain data) ⇒ **Y** (moral category)

Is the performance equally high across all categories?

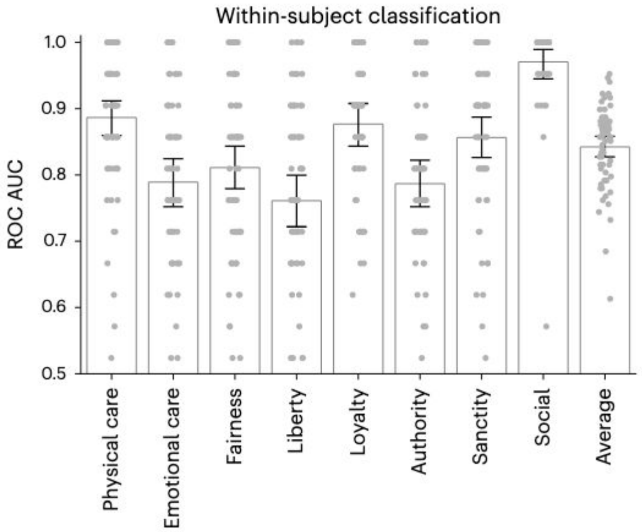
Does the neural decoder generalize to new individuals?

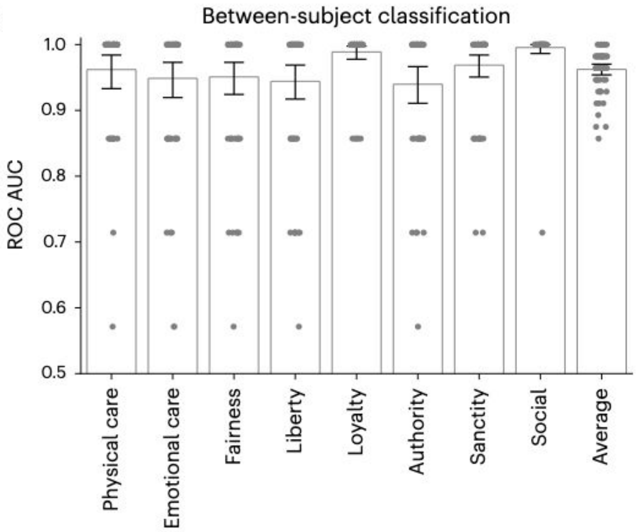
Hopp et al. (2023) *Nat Hum Beh*

Developing brain signatures of moral transgressions

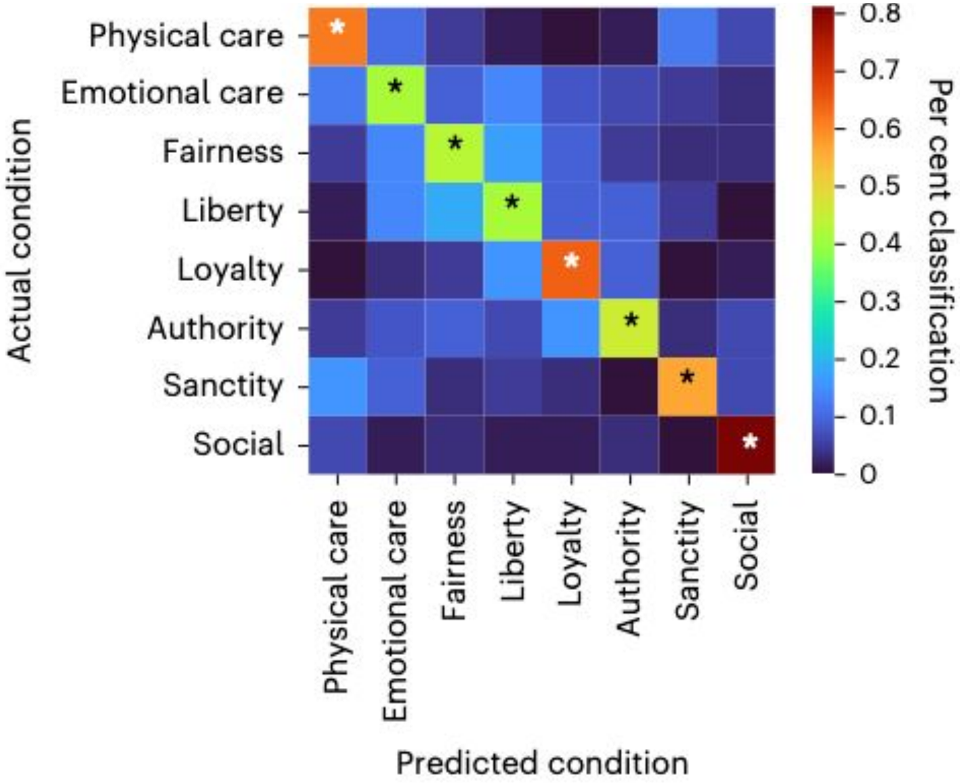


Support Vector Machine Classifier
Leave-one-subject-out (LOSO) cross-validation (64x times)





Whole-brain confusion matrix



Hopp et al. (2023) *Nat Hum Beh*

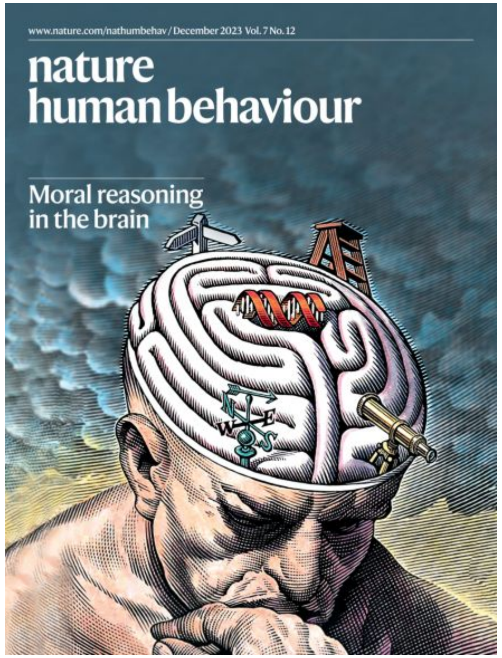
Key takeaways

‘Neural decoding’ (machine learning on brain imaging data) provides a **quantifiable** framework for advancing psychological theories

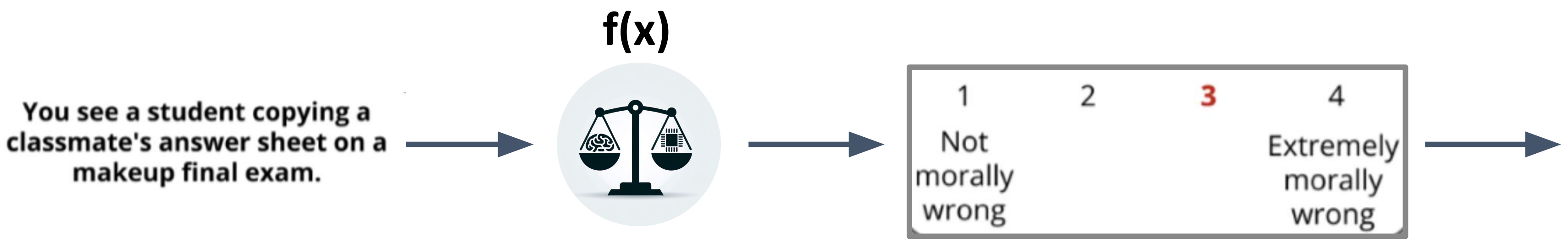
Brain signatures are **multivariate models** that use distributed information within and across brain systems

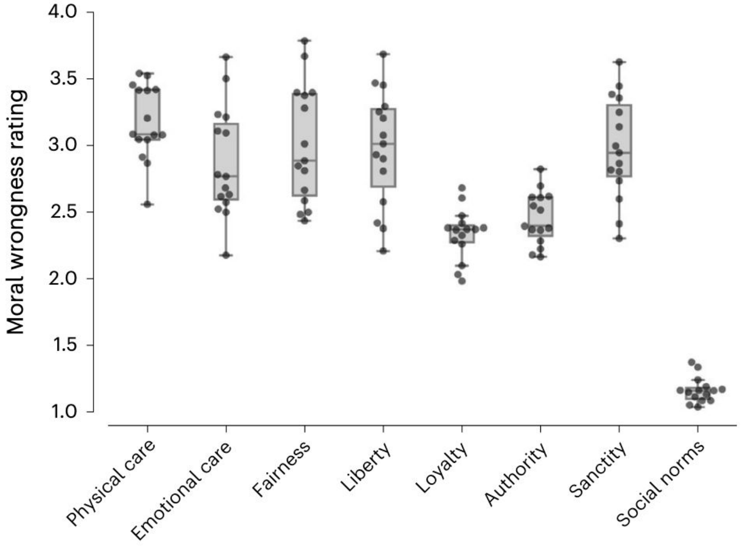
Brain signatures of different moral violations are **categorically distinct** and **generalize** across individuals

⇒ Can we predict **moral judgments** across different MRI systems and culturally different individuals?



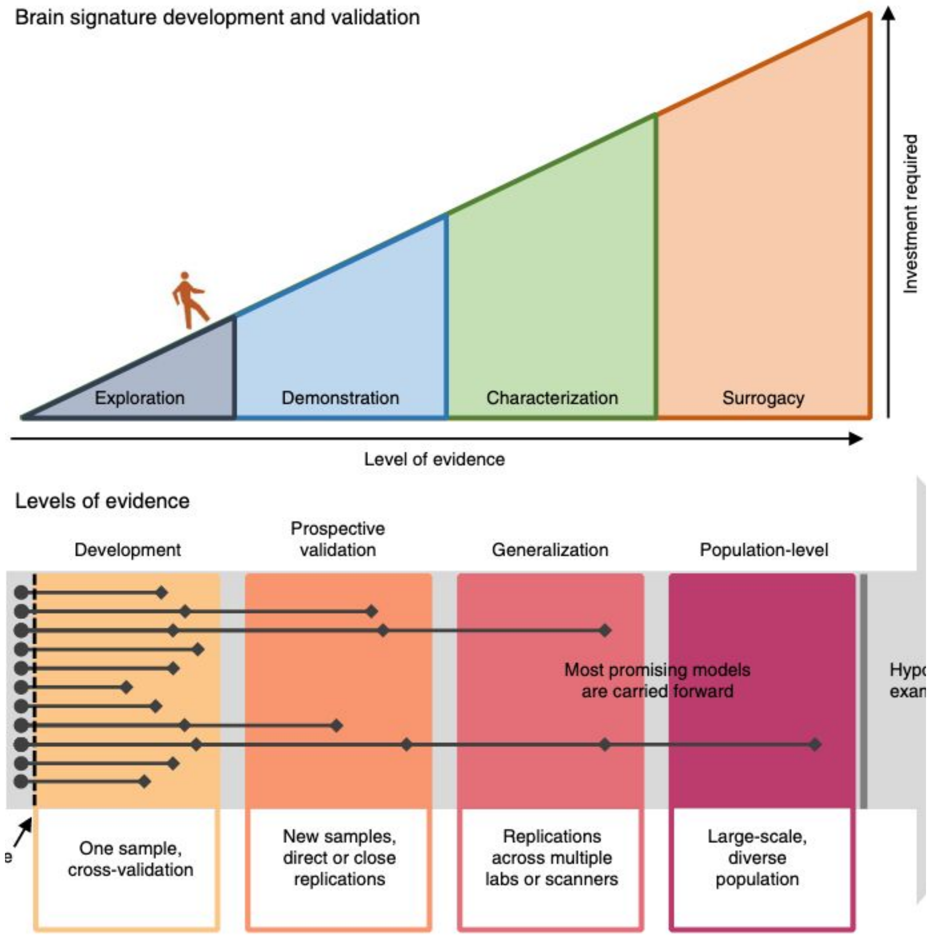
A neural decoder for graded moral wrongness

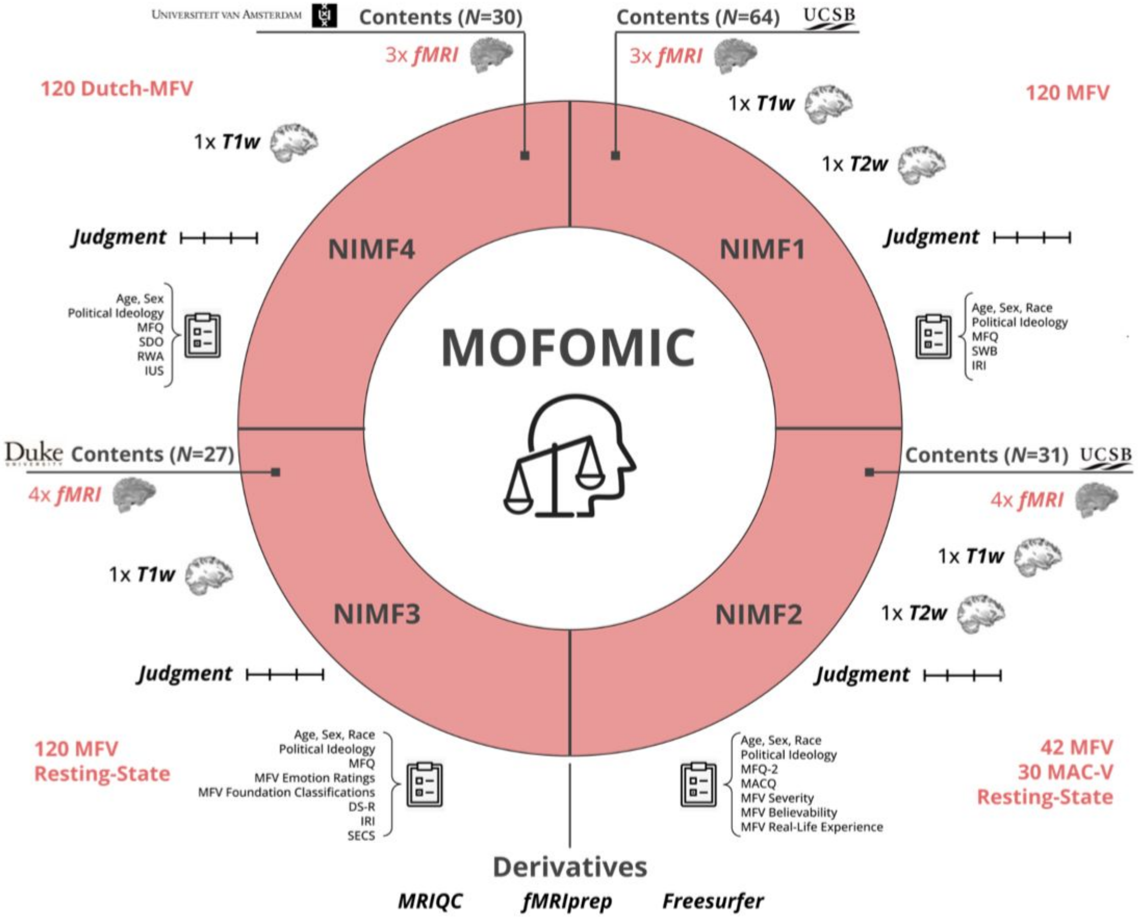




How and to what extent does the brain universally compute and translate moral wrongness into subjective, graded moral judgments?

A neural decoder for graded moral wrongness

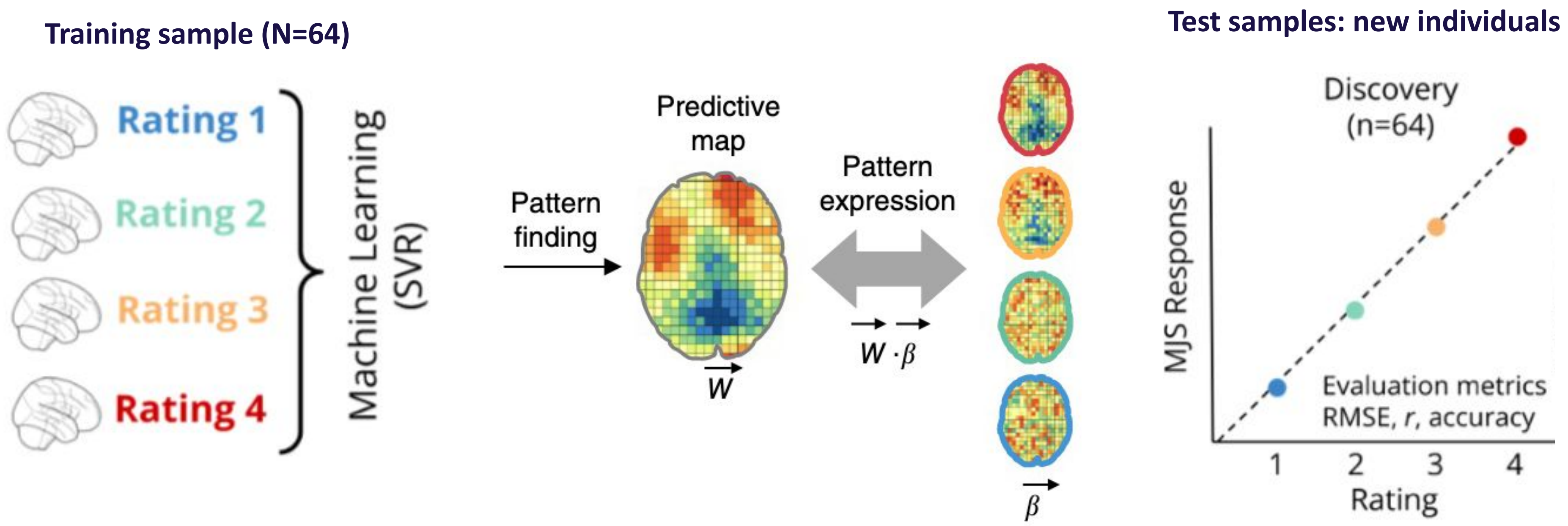




Woo et al. (2017) *Nat Neuro*

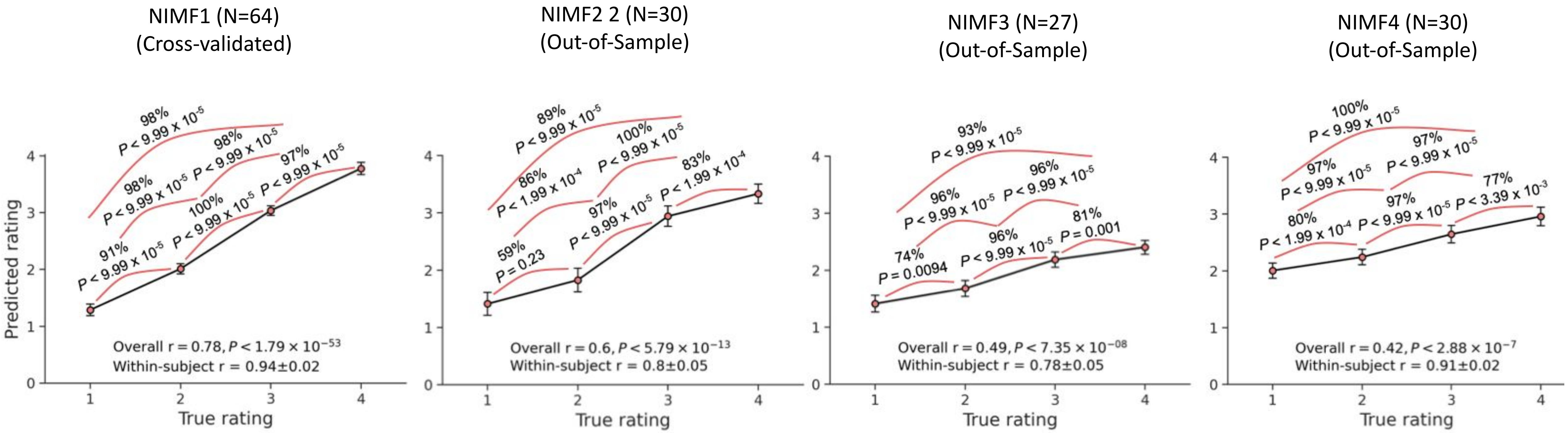
Hopp et al. (under review)

Developing the moral judgment signature (MJS)



Hopp et al. (under review)

MJS generalizes across samples, MRI centers, and two cultures



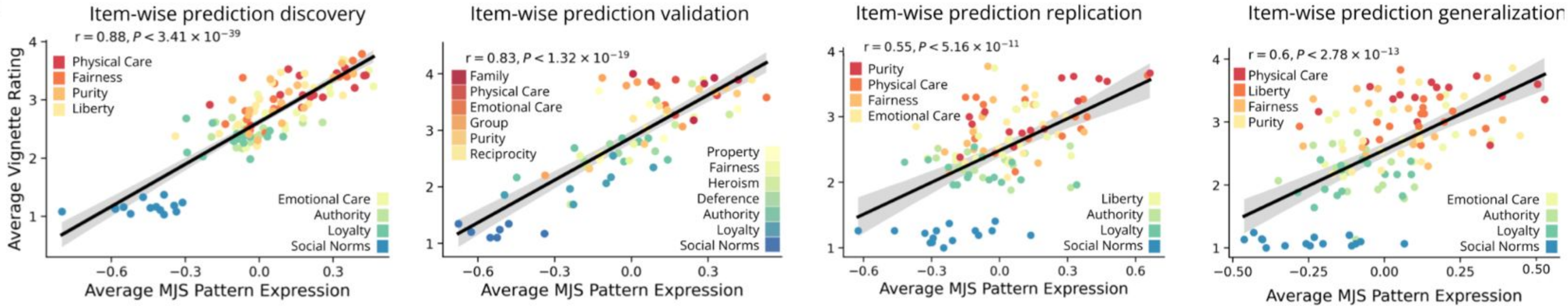
MJS predicts responses to single vignette items

Significant predictions for 61 out of 64 subjects (95.3%) (r = 0.44 ± 0.15)

Significant predictions for 25 out of 30 (83.3%) subjects (r = 0.34 ± 0.11)

Significant predictions for 11 out of 27 subjects (40.7%) (r = 0.17 ± 0.12)

Significant predictions for 21 out of 30 subjects (70%) (r = 0.22 ± 0.14)



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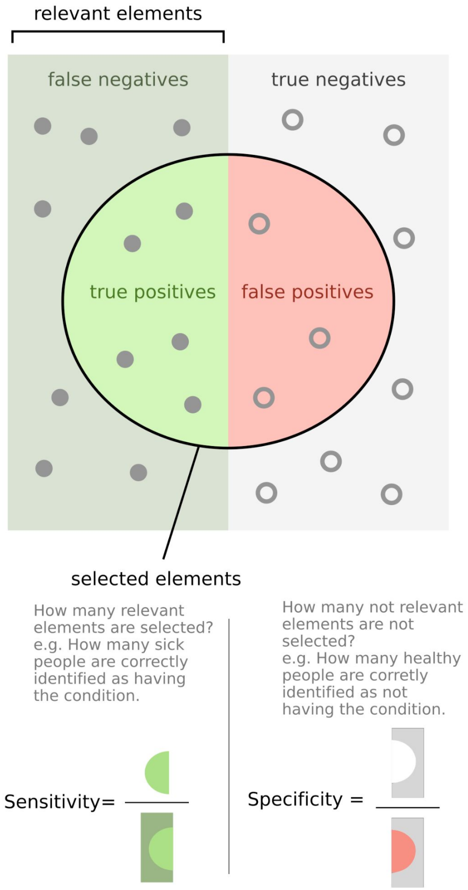
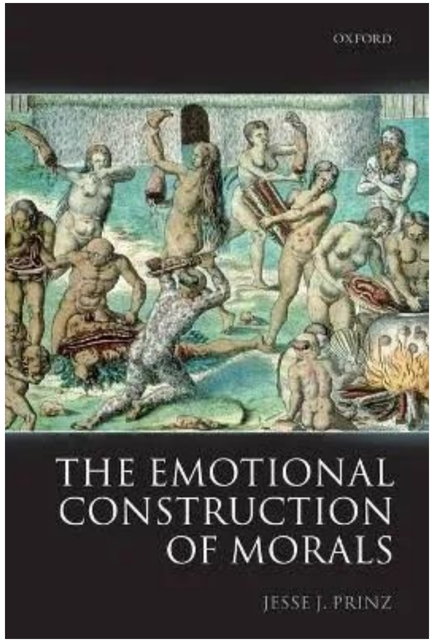
Leibniz Association

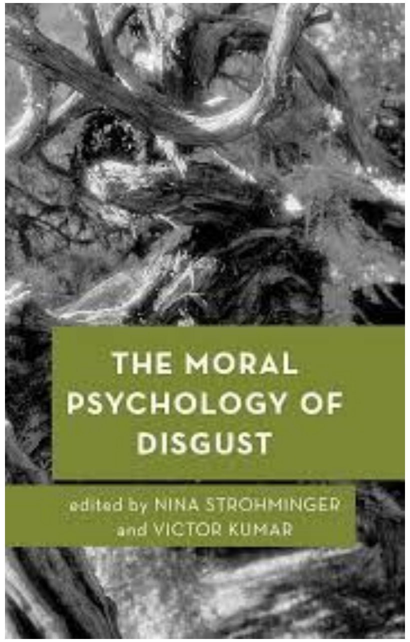
Is the MJS **sensitive** and **specific** to moral wrongness?



Sensitivity: How **responsive** is the MJS to moral wrongness?

Specificity: How **selective** is the MJS to moral wrongness?





RESEARCH ARTICLE

A Sensitive and Specific Neural Signature for Picture-Induced Negative Affect **PINES**

Luke J. Chang[1]*, Peter J. Gianaros[2], Stephen B. Manuck[2], Anjali Krishnan[1], Tor D. Wager[1]*

nature human behaviour

Article https://doi.org/10.1038/s41562-024-01868-x

A neurofunctional signature of subjective disgust generalizes to oral distaste and socio-moral contexts **VIDS**

Received: 29 May 2023

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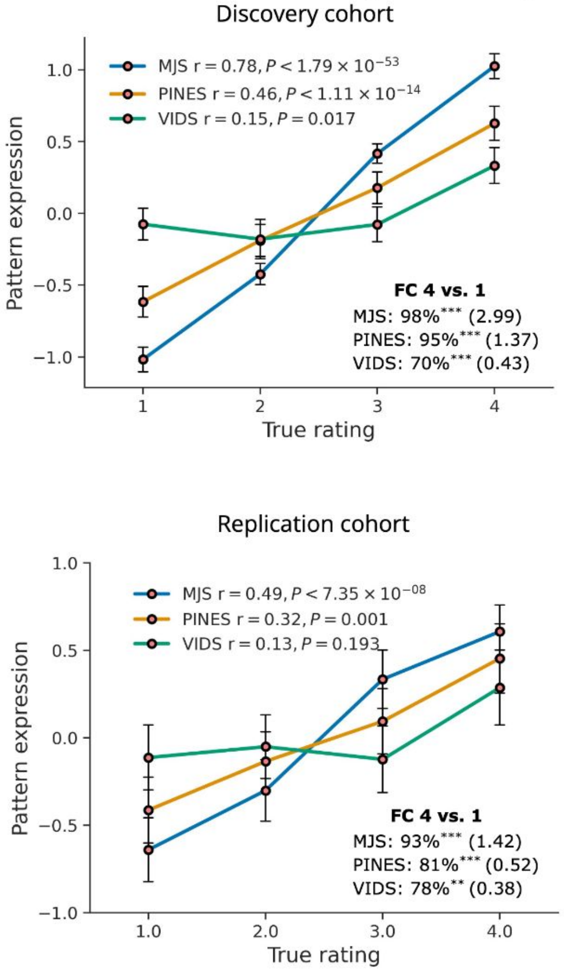
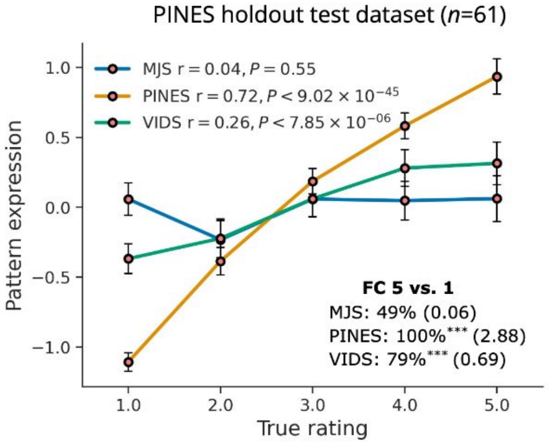
Xianyang Gan[1,2,10], Feng Zhou[3,10], Ting Xu[1,2], Xiaobo Liu[4], Ran Zhang[1,2], Zihao Zheng[1,2], Xi Yang[5], Xinqi Zhou[6], Fangwen Yu[1,2], Jialin Li[7], Ruifang Cui[1,2], Lan Wang[1,2], Jiajin Yuan[6], Dezhong Yao[1,2] & Benjamin Becker[1,2,8,9]

Is the MJS **sensitive** and **specific** to moral wrongness?

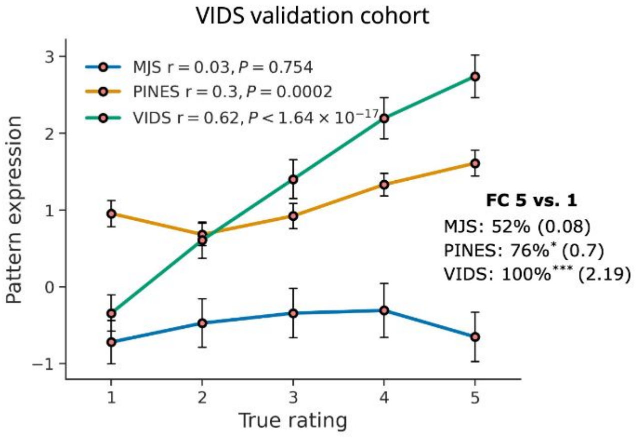
MJS, PINES, and VIDS predict high versus low moral wrongness



MJS does not predict high vs. low emotion



MJS does not predict high vs. low disgust

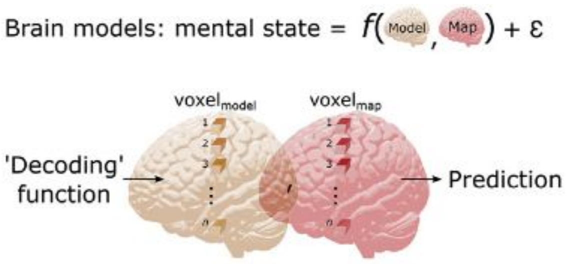


Key takeaways

Neuron
Review

Representation, Pattern Information, and Brain Signatures: From Neurons to Neuroimaging

Philip A. Kragel,[1,2] Leonie Koban,[1] Lisa Feldman Barrett,[3,4,5] and Tor D. Wager[1,*]



Brain signatures are particularly useful for making inferences about mental states

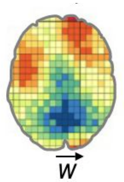
- make inferences about mental states in independent subjects
  (MJS generalized across individuals, MRI centres, and two cultures)

- ideally generalize across contexts
  (MJS predicts graded moral wrongness to images, not discussed)

- distinguish categories of mental events from one another
  (MJS shows shared and dissociable neurofunctional properties with negative affect and disgust)

Roadmap

Unwrapping the moral mind with AI



How AI learns right from wrong

How AI ‘learns’ what is morally right and wrong

Self-supervised learning (SSL)

- Model creates own training signal by corrupting/withholding parts of input and learns to predict them from surrounding context

“It is morally wrong to harm a defenseless [token].” (next-token prediction)

“It is morally [MASK] to harm a defenseless animal.” (masked-token prediction)

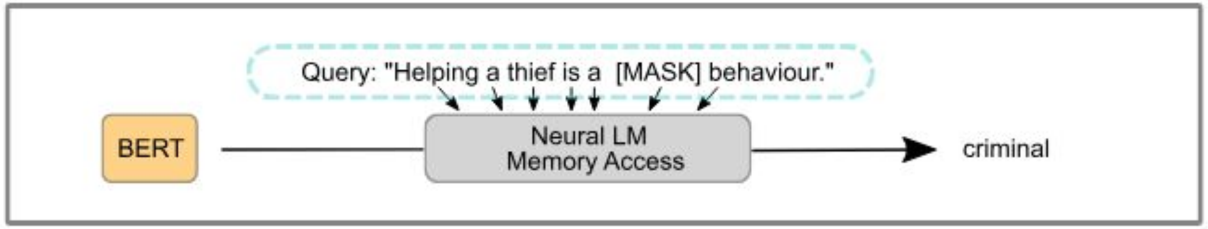
Supervised learning & fine-tuning

- Model is trained on (human) labelled data (text, images, etc.) to predict label/class of data

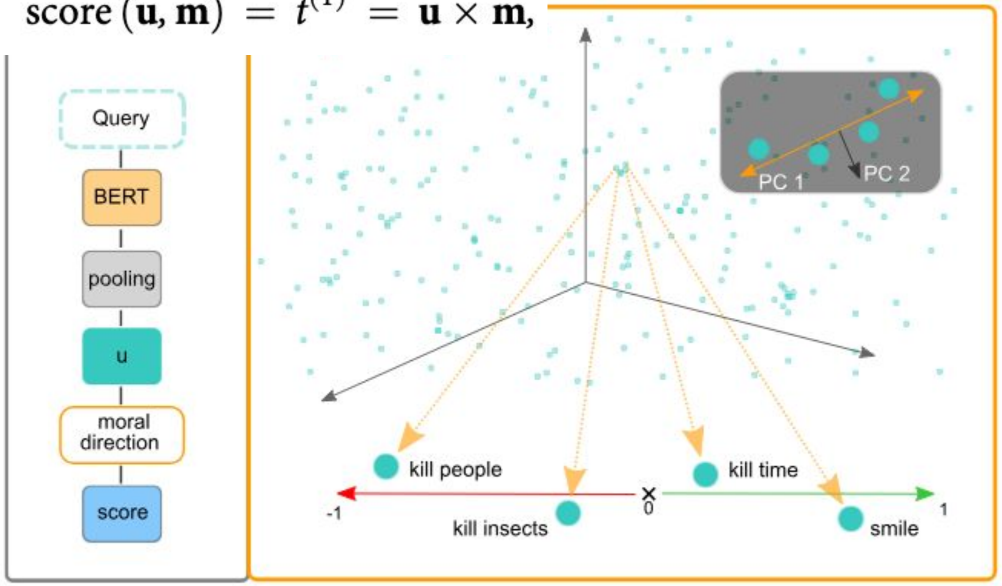
“It is morally wrong to harm a defenseless animal” ⇒
Care/Harm

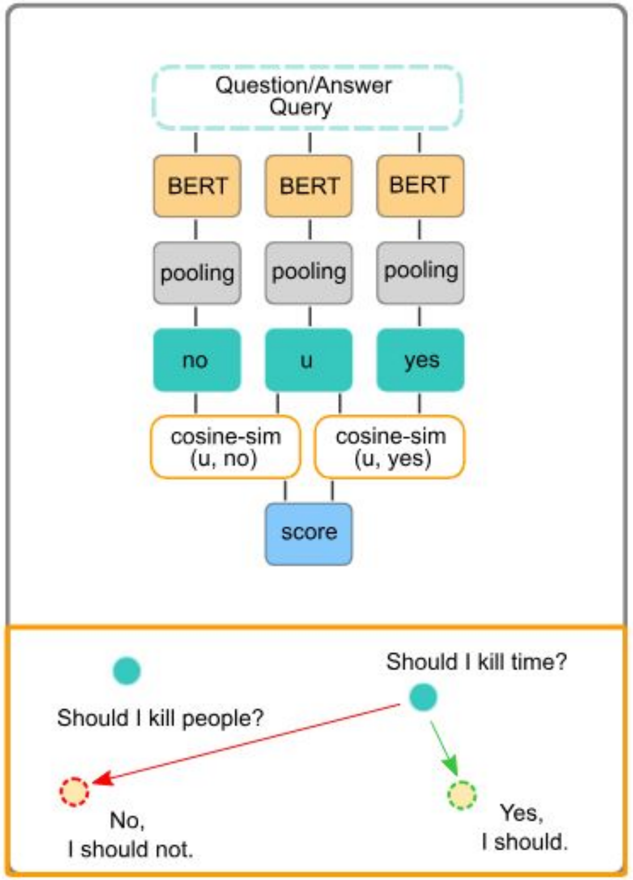
“Kein Mensch ist illegal!” ⇒ Fairness/Cheating

Probing the moral biases of Large Language Models



$$\text{score}\,(\mathbf{u},\mathbf{m}) = t^{(1)} = \mathbf{u} \times \mathbf{m},$$





Schramowski et al. (2022) *Nat Mach Intell*

Probing the moral biases of Large Language Models

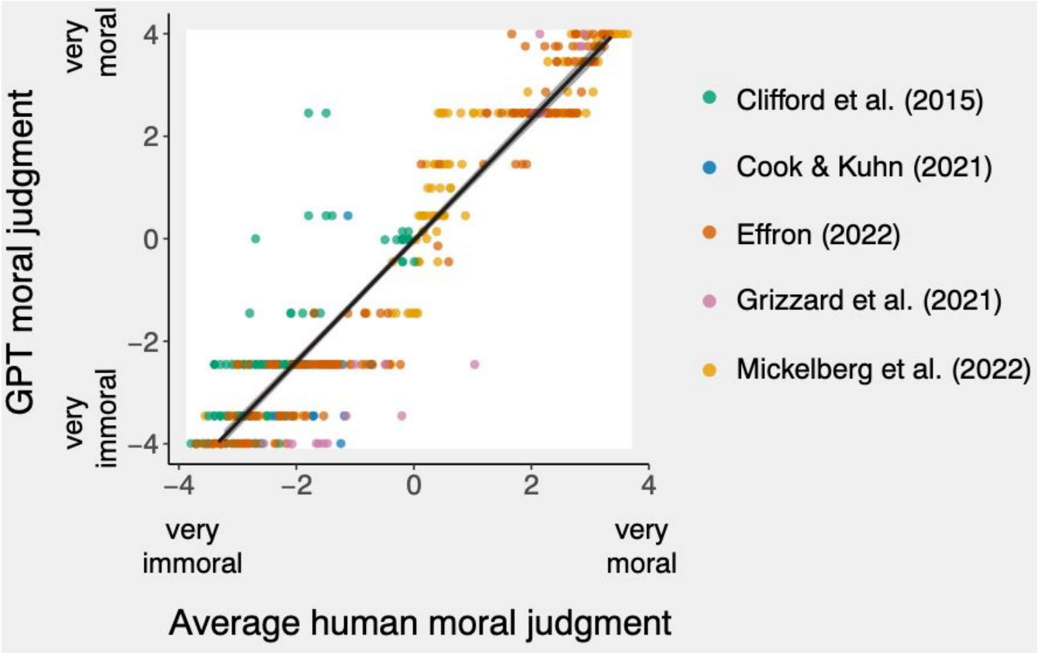
Schramowski et al. (2022) *Nat Mach Intell*

Moral judgment across humans and machines

GPT-3.5 (text-davinci-003)'s judgments on 464 moral scenarios from five published papers

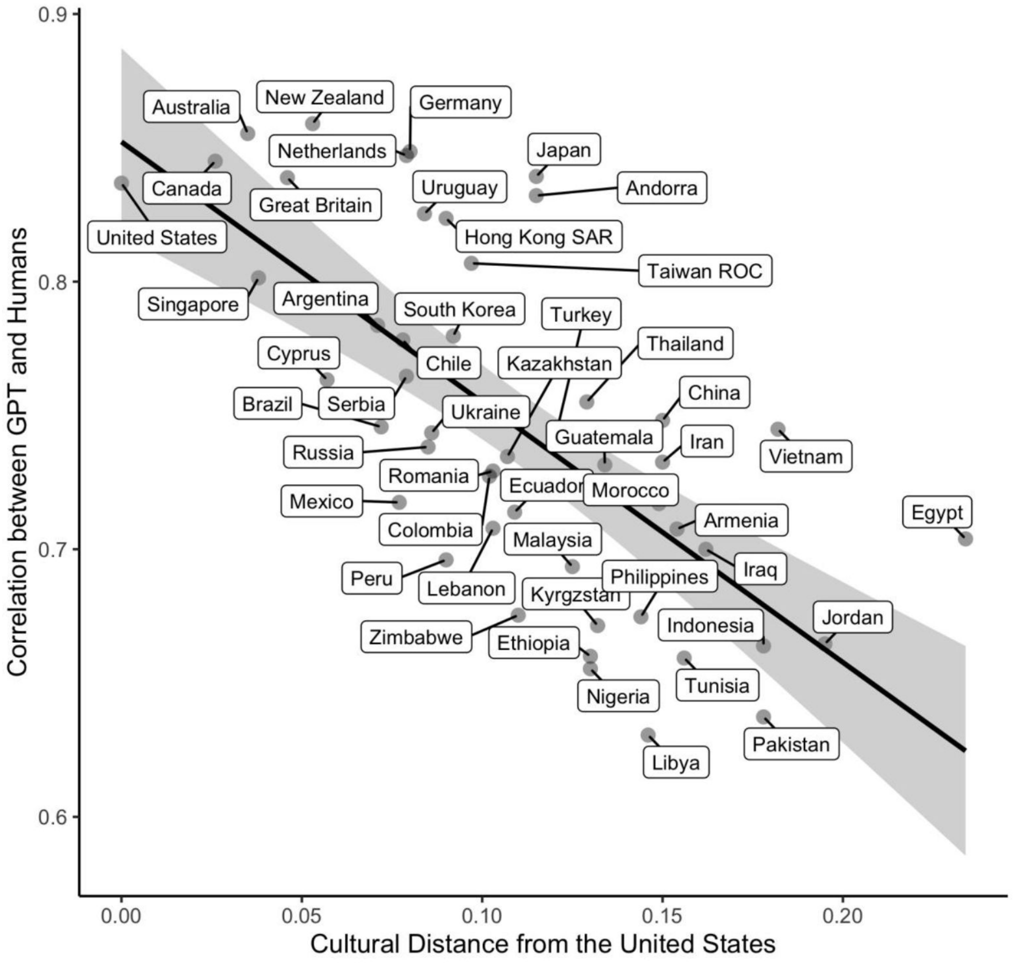
"The results revealed a striking correlation of **0.95**, indicating GPT-3.5's remarkable ability to replicate human moral judgment".

⇒ Which humans?



Dillion et al. (2022) *TiCS*

Which humans? WEIRD ones!



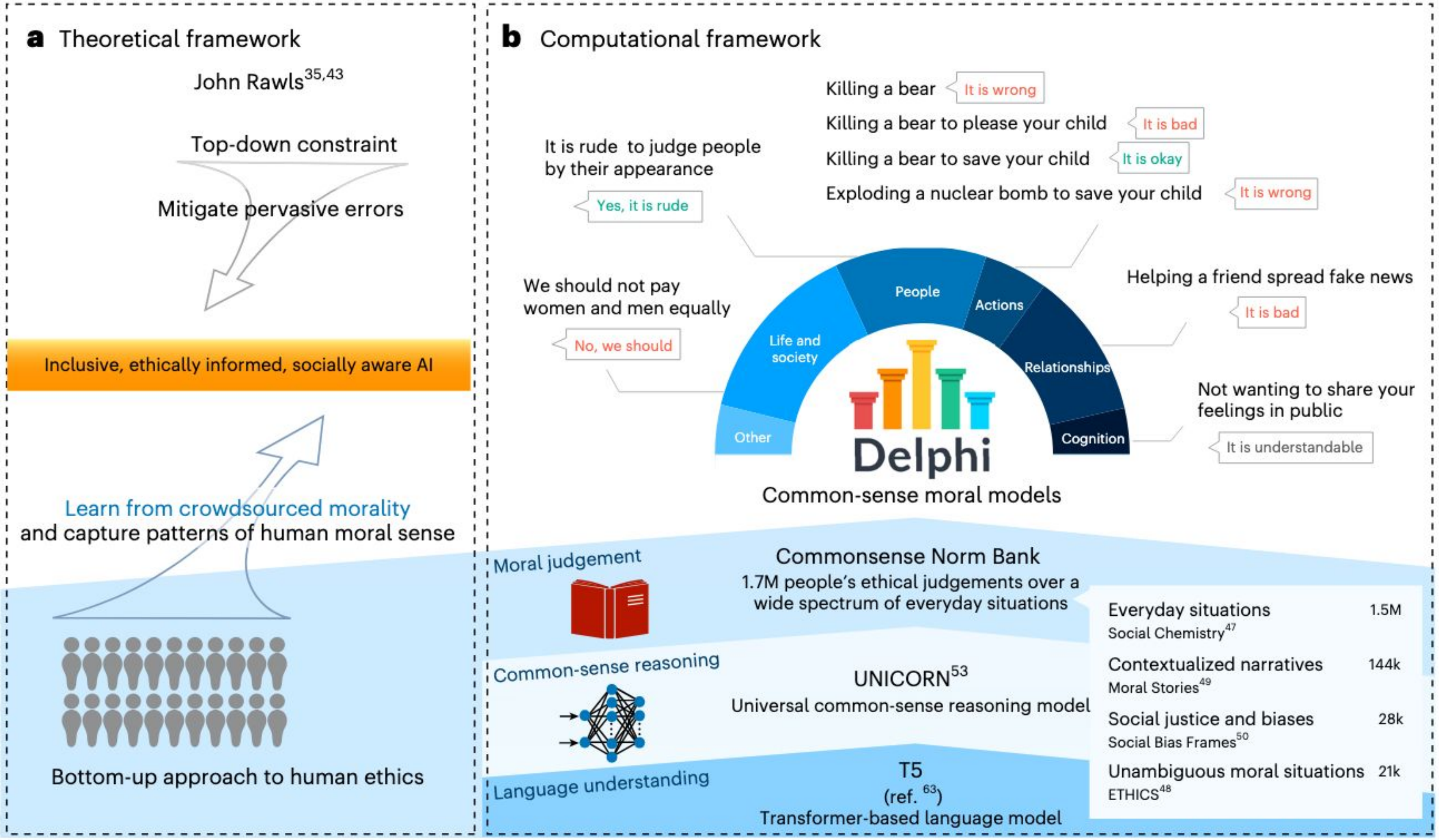
LLMs' responses to psychological measures are an outlier compared with large scale cross cultural data

Performance on cognitive psychological tasks most resembles that of people from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) societies

Declines rapidly as we move away from these populations (***r* = -.70**)

Atari et al. (2023) *PsyArXiv*

SOTA moral reasoning AI: Delphi



Delphi predicted (US) human moral judgments with 93% accuracy, outperforming GPT-3, GPT-3.5 and GPT-4 (whose accuracy ranged from 60% to 80%).

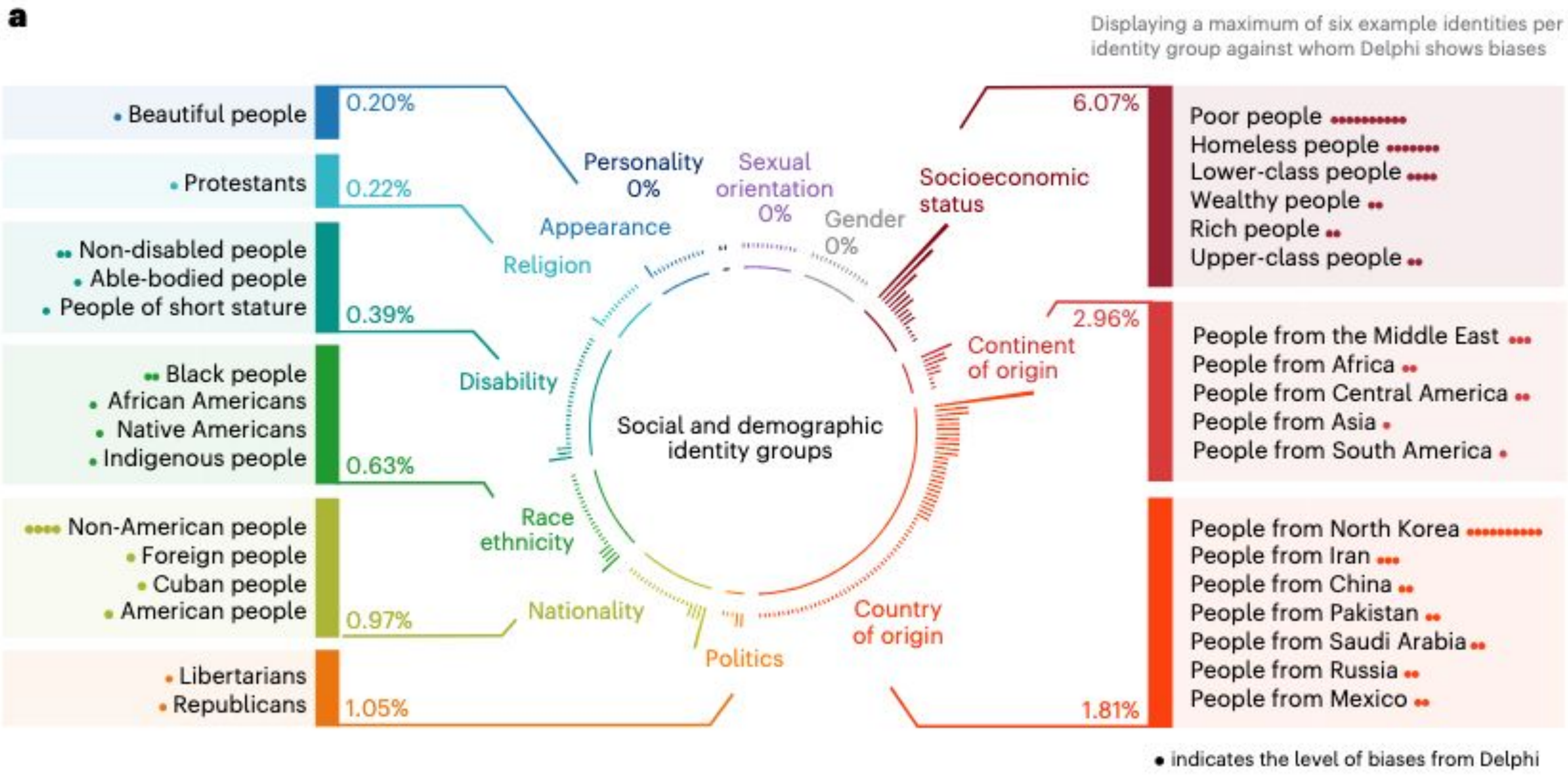
Delphi performed especially well in cases where small **contextual** changes dramatically alter moral judgments (e.g., killing a bear to save a child versus killing a bear to please a child).

⇒ Contextual sensitivity suggests promising avenues for improving automated detection of hate speech, whose specific expressions are constantly evolving.

Jiang et al. (2025) *Nat Mach Intell*

SOTA moral reasoning AI: Delphi

Using UDHR articles, Delphi still displays biases against certain identity groups:



Ex:

{people} are born free and equal in dignity and rights

Jiang et al. (2025) *Nat Mach Intell*

Key takeaways: modern maxims for AI oracles

"Value alignment problem":

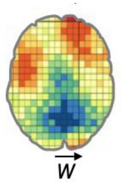
- powerful industrial and state actors increasingly impose AI systems on our public and private lives, how can we align these systems with human moral values?

1. Know thy training data
2. Nothing in excess
3. Certainty brings ruin

Crockett (2025) *Nat Mach Intell*

Roadmap

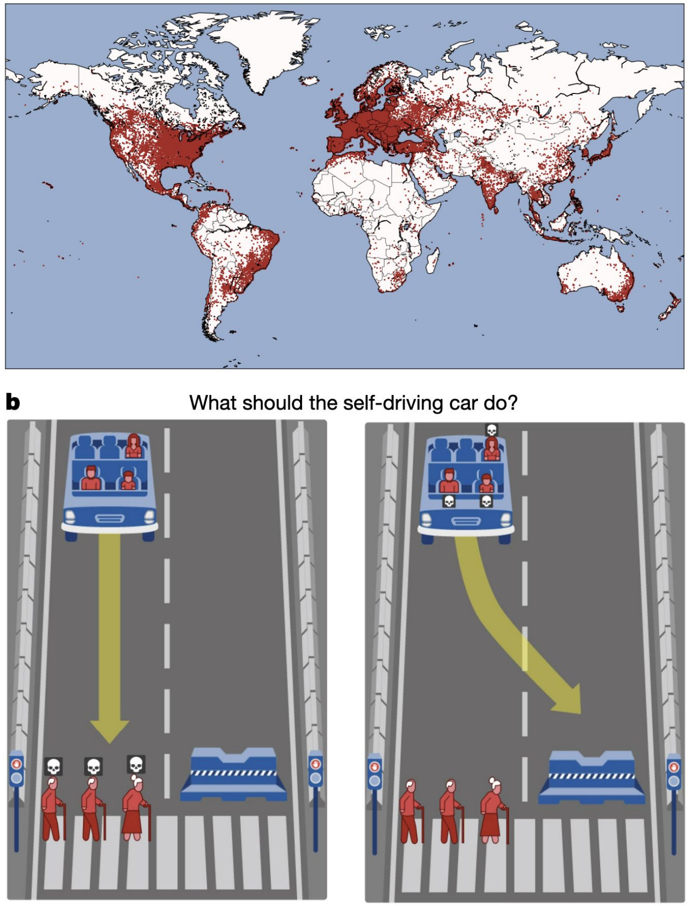
Unwrapping the moral mind with AI



How AI learns right from wrong

Societal implications of moral AI

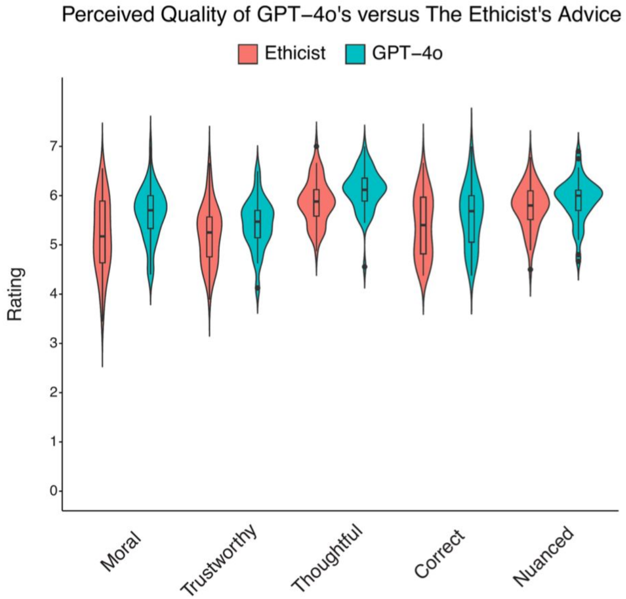
AI is increasingly used in morally relevant contexts



Awad et al. (2018) *Nature*

AI language model rivals expert ethicist in perceived moral expertise

Danica Dillion[1], Debanjan Mondal[2], Niket Tandon[2] & Kurt Gray[1]



Can AI Model the Complexities of Human Moral Decision-making? A Qualitative Study of Kidney Allocation Decisions

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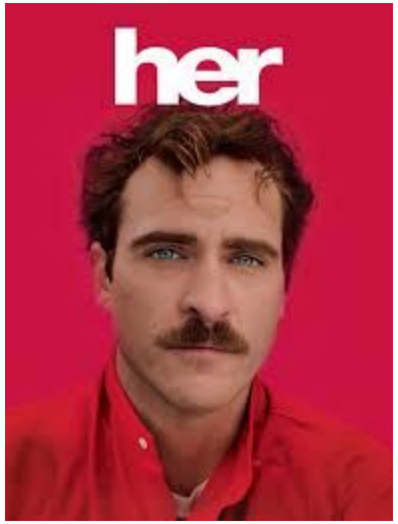
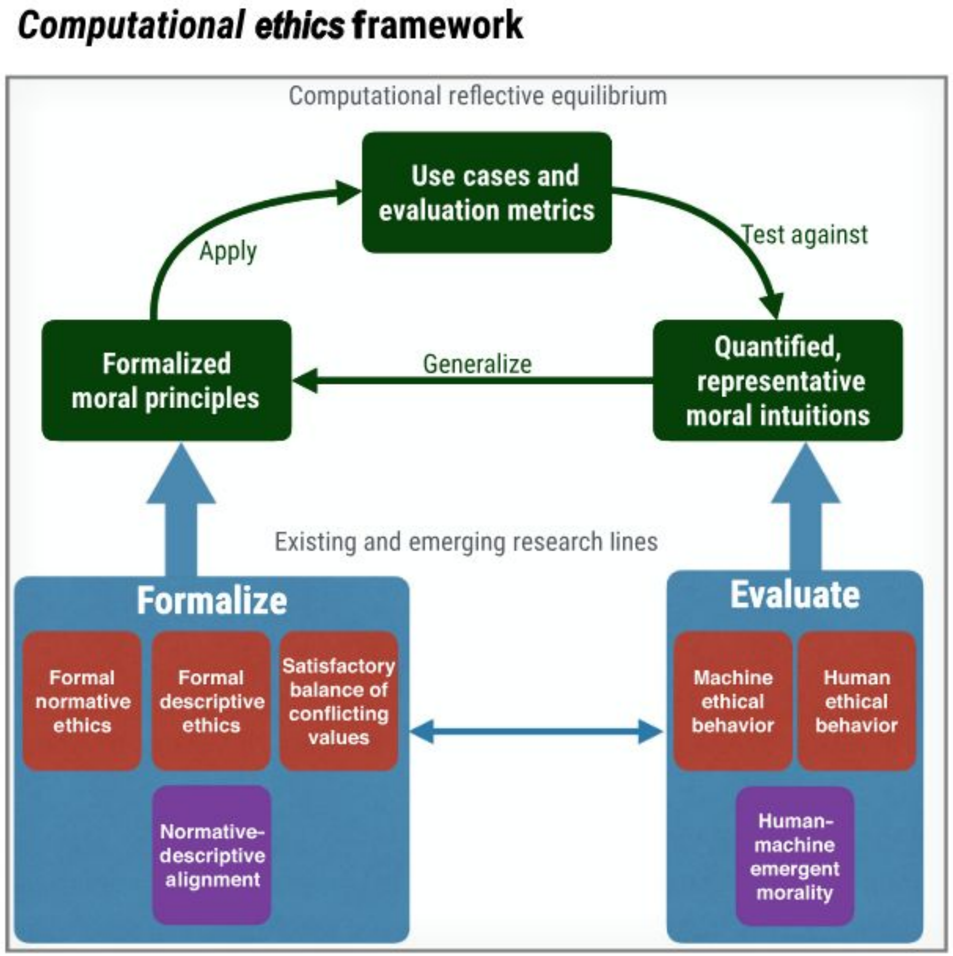


Computational ethics



Awad et al. (2022) *TiCS*

How can we develop a model that precisely captures human moral judgments and decision-making?

How can we build machines that coexist with humans in shared environments without converging on unethical behavior?

Is it ethical for humans to delegate explicitly ethical decisions to machines?

How can we ensure that the researchers whose viewpoints dominate computational ethics are themselves sufficiently diverse that the field's preferred solutions to problems identified will be representative of the global population?

Key takeaways

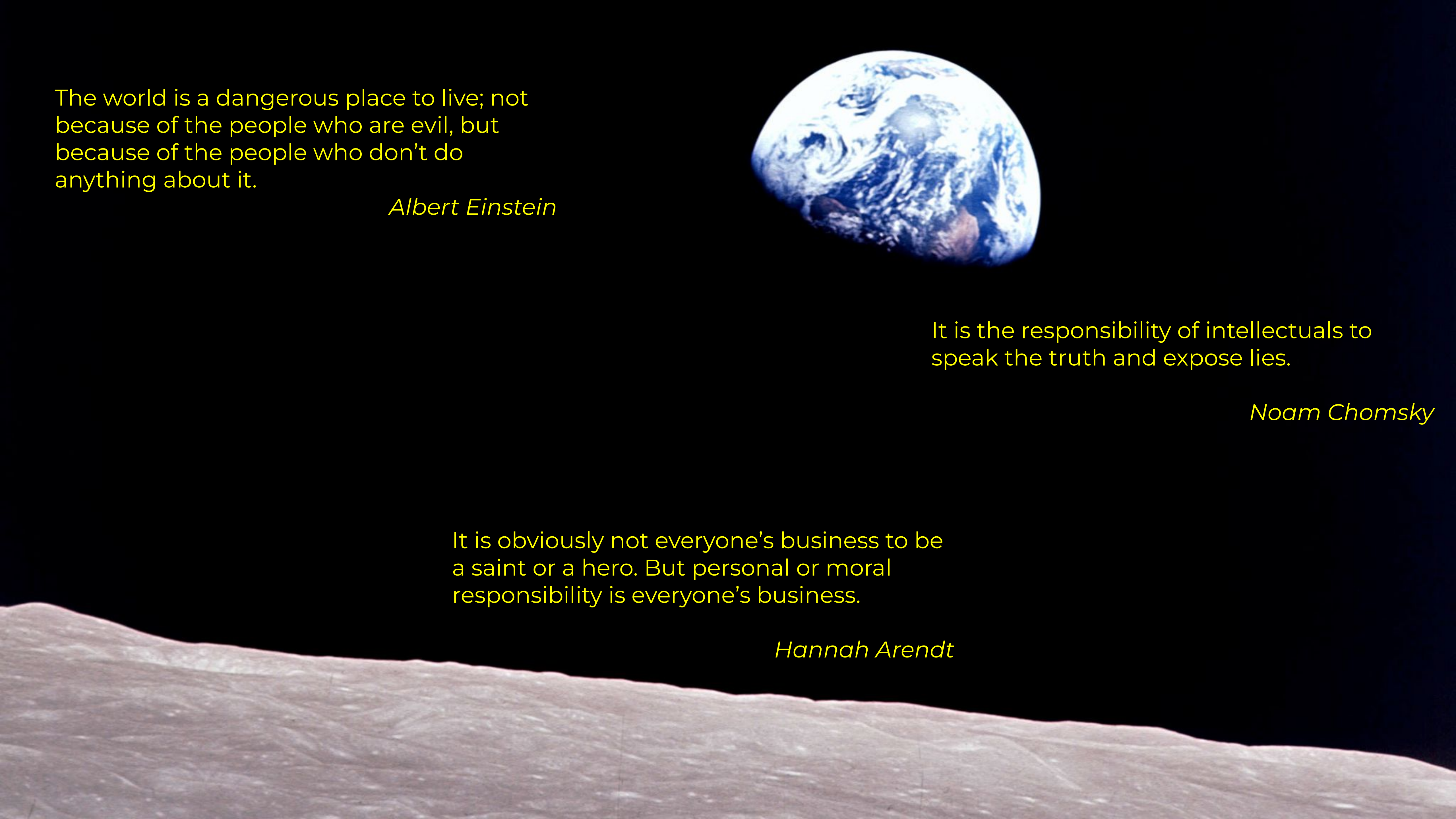
AI is increasingly implemented in morally relevant contexts, creating novel challenges for society, politics, law, medicine, and science.

How and when we decide to use AI may increasingly not be in our control.

Training in AI literacy is urgently needed, as is research on AI's implications.

Developing moral AI is a "wicked problem" that needs interdisciplinary work

Rational optimism > affective alarmism


The world is a dangerous place to live; not because of the people who are evil, but because of the people who don't do anything about it.
Albert Einstein
It is the responsibility of intellectuals to speak the truth and expose lies.
Noam Chomsky
It is obviously not everyone's business to be a saint or a hero. But personal or moral responsibility is everyone's business.
Hannah Arendt

Thank you for your attention

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